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Industrialization vs Decarbonization: A Data-Driven Analysis of Major European Economies

Data Analytics / Explainer Project

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Abstract

Carbon prices in European Union (EU) have surged to record highs, raising a critical economic question: are we actually reducing emissions, or are we just closing down factories? Distinguishing between true "decarbonization" (becoming more efficient) and "deindustrialization" (losing our manufacturing base) is vital for Europe's economic future. To find the answer, this project analyzes industrial trends across six EU countries from 2010 to 2025. We constructed a unified dataset that combines daily carbon prices (ICE), factory output numbers (Eurostat), and total emissions data (EDGAR). Using a statistical model (fixed-effects regression), we isolated the specific impact of carbon taxes on industry, separating it from other factors like the 2022 inflation crisis. This analysis aims to prove whether high carbon prices are successfully driving green innovation or simply hurting industrial competitiveness.

Keywords: Greenhouse Gases (GHGs), Carbon Tax, Carbon Pricing, European Union (EU) Emission Trading System (ETS), Industrialization, Decarbonization, Global Warming

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List of Abbreviations

EU	European Union	H₂O	Water Vapor
GHG	Greenhouse Gas	SO₂	Sulfur dioxide
GHGs	Greenhouse Gases	CO₂	Carbon dioxide
ETS	Emission Trading System	CH₄	Methane
UNFCCC)	United Nations Framework Convention on Climate Change	SF₆	sulphurhexafluoride
EUA	European Union Allowance	N₂O	Nitrous oxide
EDGAR	Emissions Database for Global Atmospheric Research	GVA	Gross Value Added
EUTL	European Union Transaction Log	FE	Fixed Effects
FE	Fixed Effects (Statistical Model)		
ICE	Intercontinental Exchange (Source for carbon futures data)		
NACE	Statistical Classification of Economic Activities in the European Community		
PH	Porter Hypothesis (Innovation theory)		
PHH	Pollution Haven Hypothesis (Carbon leakage theory)		
PPI	Producer Price Index (Measure of inflation)		
HFCs	Hydro Fluorocarbons		
PFCs	Per Fluorocarbons		
R&D	Research and Development		

1 Introduction

In today's modern world, we often hear the terms like climate changes, global warming, rise of sea level due to melting of polar ice caps [9] and much more that sound like the planet is about to be destroyed, but we do not know why these are a big topic of discussion across countries around the world. In a nutshell, all these terms point out to a concept of Greenhouse Gases or GHGs and its concentration in the environment. GHGs are the naturally occurring atmospheric gases that are present around us and the main purpose of its existence is to trap the amount of heat that radiates from the surface of the earth and maintain the warmth of the planet so that life can be able to sustain in the world. This GHGs consists of handful of gases such as the carbon dioxide (CO_2), sulfur dioxide (SO_2), nitrous oxide (NO_2), methane (CH_4), water vapor (H_2O), and some types of gases that contain fluoride in them (F-gases). Although these gases are already present in the atmosphere, they can also be produced by human activities such as industrialization, burning of fossil fuels, agriculture, deforestation and so on. When the GHGs are emitted in large quantities, they negatively affect the planet by causing chemical imbalance of their composition in the environment. Therefore, GHGs are responsible for the rise of global temperature of the earth and because of this change in temperature, the living beings on earth will be affected [4].

The most amount of GHGs are produced by the burning of fossil fuels and industrialization, and out of all the gases in the GHG family, carbon dioxide (CO_2) is the most largely emitted gas because these processes easily emit vast quantities of CO_2 compared to other gases and because of this, carbon is added faster in the atmosphere than natural processes like oceanic absorption and photosynthesis can remove it, and as a result, the carbon cycle of the earth becomes unbalanced. Thus in the end, out of all the gases from the GHG family, carbon will be present in huge amounts and will also stay for a very long period of time compared to other gases, and because of this critical reason, carbon is the most important gas that is to be controlled.

In the early stages of industrialization, the only means of energy source was the fossil fuels. They have to be burned in order to produce energy for the industries to function. And because there is no control over the consumption of fossil fuels, the carbon emission, along with other GHGs was very enormous and due to this emission the average world temperature started to increase by 1.4°C to 5.8°C every year [9]. Considering this fact, a measure of controlling carbon emission has to be established, which is the concept of Carbon Taxing or Carbon Pricing that was ini-

tially proposed by Arthur Pigou in the year 1932 that was also called the Pigouvian Tax, and this was firstly introduced in Finland in the year 1990 [11]. A carbon tax is a type of payment or fee that is directly applied on the carbon emissions with a regulated marginal price [11]. Being a cost-controlling tool, this carbon tax rate has been widely studied for the optimal application [11]. After the introduction of carbon tax policy, European Union (EU) countries such as Sweden, Denmark and other Nordic countries have started to adopt them, and in 1992, the United Nations Framework Convention on Climate Change (UNFCCC) was adopted which was one of the first international measures to address this problem of climate change by stabilizing the concentration of GHGs in the atmosphere, and this came into effect in the year 1994 [5].

Then in the year 1994, the governments around the world were building on the UNFCCC framework and held a meeting in Kyoto where they have adopted a protocol which stated that the developed countries and the countries in transition in their economies would reduce their emission of the six major GHGs namely carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), hydro fluorocarbons (HFCs), per fluorocarbons (PFCs), and sulphurhexafluoride (SF_6) [10], by at least 5% below their levels in the year 1990 [5], and this protocol was named the Kyoto Protocol and this came into effect in February 2005 [5]. The timeframe for this reduction is meant to be a five-year period that is between 2008 and 2012 [5]. In the case of Europe, they produced around 14% of the total world's GHGs and under the Kyoto Protocol, the EU have agreed to cut the emissions by 8% during this allotted period and the levels of reduction were not uniform for all of the countries [5]. Therefore, the first approach of the EU was to decide on how the cuts have to be distributed, and then they created emission allowances that allow their owners to emit a certain quantity of CO_2 , and this lead to the establishment of a trading mechanism called the EU Emission Trading System (ETS) in the year 2005 [5]. In general, the ETS is a platform to provide economic incentives for carbon reduction of the participating countries [11]. The EU ETS operates with a "cap-and-trade system" in which, the emission allowance for each pollutant is determined in advance by the regulation in accordance with the CO_2 reduction goal [11].

After establishment of the EU ETS, they became the world's largest cap-and-trade program and arguably the most important market-based application of the economic principles to the climate change problems [3]. Also, the EU currently stands at the forefront of the most significant economic transformation since the industrial revolution. With the "Fit for 55" legislative package, the bloc has legally bound itself to reduce net GHG emissions by at least 55% by 2030. To achieve this, the EU relies primarily on the concept of Carbon Pricing, which is a market-based mechanism designed to monetize the external cost of pollution. This system, anchored by the EU ETS and supplemented by national carbon taxes, aims to force a transition from fossil fuels to renewable energy. However, this transition has created a critical economic tension; **Industrialization vs. Decarbonization**.

While the environmental necessity of reducing CO₂ is undisputed, the economic consequences are heavily debated. On one side, the "Porter Hypothesis" suggests that stringent carbon pricing acts as a catalyst, pushing companies to innovate, adopt green technologies, and ultimately become more productive. On the other side, concerns regarding "Carbon Leakage" and "Greenflation" suggest that high energy costs may erode the competitiveness of European heavy industry (steel, chemicals, manufacturing), forcing production to move offshore to countries with laxer regulations. This tension is no longer theoretical. Recent data from 2023-2025 indicates economic stagnation in key industrial hubs like Germany, raising the question: *Is this the temporary cost of transition, or the beginning of structural deindustrialization?*

This project investigates this dilemma by analyzing data from the EU-6 (Germany, France, Italy, Poland, Spain, Netherlands) between 2010 and 2025. By combining carbon pricing data with macroeconomic indicators and innovation metrics, this report aims to determine whether the EU ETS is a driver of green growth or a brake on industrial competitiveness.

2 Goals & Milestones

The primary objective of this research is to answer the core question: **Does stringent carbon pricing in the EU ETS reduce industrial competitiveness or drive green innovation?** To answer this, the project is structured around three specific research goals and a phased execution plan.

2.1 Research Goals

Goal 1: Quantify the "Regulatory Drag" (Competitiveness Analysis)

- **Objective:** To measure the correlation between rising Carbon Prices (EU ETS) and Industrial Output (Gross Value Added) across the EU-7.
- **Hypothesis to Test:** We aim to determine whether high carbon prices cause a statistically significant contraction in energy-intensive sectors (e.g., Manufacturing, Steel) relative to less energy-intensive sectors.

Goal 2: Test the "Porter Hypothesis" (Innovation Analysis)

- **Objective:** To track whether increases in carbon pricing lead to a subsequent rise in "Green Patents" and R&D efficiency.
- **Hypothesis to Test:** We aim to identify if there is a "lag effect" where high prices today result in increased technological innovation 2-3 years later.

Goal 3: Assess the "Valley of Death" (Transitional Risk)

- **Objective:** To analyze the specific period of 2021–2025, where carbon prices and energy inflation spiked simultaneously.
- **Hypothesis to Test:** We aim to determine whether the combination of high carbon taxes and high energy prices creates a "double burden" that outweighs the short-term benefits of innovation.

2.2 Project Milestones

The project is divided into three sequential phases to ensure a logical flow from theory to data evidence.

Phase 1: Foundation & Theory

- **M1.1 Literature Review:** Complete a review of key economic theories, explicitly contrasting the *Porter Hypothesis* (Innovation) against the *Pollution Haven Hypothesis* (Leakage).
- **M1.2 Theoretical Framework:** Develop a causal diagram explaining how Carbon Pricing transmits through the economy to affect GDP and Innovation.

Phase 2: Data Architecture

- **M2.1 Data Collection:** Acquire raw datasets for Carbon Prices (Ember/ICE), Economic Output (Eurostat), Emissions (EDGAR/EUTL), and Innovation (OECD).
- **M2.2 Data Harmonization:** Clean and merge disparate datasets into a unified "Panel Dataset." This includes harmonizing currencies to €2015 (Real terms) and matching country codes.
- **M2.3 Variable Construction:** Create the "Carbon Intensity" variable to measure the effective tax burden on different industries.

Phase 3: Analysis & Visualization

- **M3.1 Statistical Analysis:** Perform regression analysis (Fixed Effects Model) to isolate the impact of carbon pricing from other economic factors like inflation.
- **M3.2 Visualization:** Generate a suite of 10-15 graphs (Time-series, Scatter plots, Heatmaps) that visually prove or disprove the link between pricing and industrial decline.
- **M3.3 Final Reporting:** Synthesize findings into a final report that provides actionable recommendations for EU policymakers.

3 Literature Review & Theoretical Framework

Since EU ETS Phase 1 began in 2005 and up to the Market Stability Reserve reforms in 2019, one main question has persisted: Does carbon pricing during these periods harm economic performance by increasing costs and reducing competitiveness?

Previous research has helped explain the theory and gather evidence from the EU ETS. This study also develops a theoretical framework to address the identified research gap.

3.1 Theoretical Foundation

Traditional environmental economics, based on Pigou's analysis of externalities, sees environmental regulation mostly as a cost. In the case of carbon pricing, this means that making firms pay for the social cost of carbon dioxide emissions increases production costs, which can lower output, reduce jobs, and harm international competitiveness. This view is especially important when discussing energy-intensive, trade-exposed industries such as steel, cement, and chemicals [7].

Porter and van der Linde (1995) challenged the typical focus on costs. They argued that good environmental regulation can drive innovation that partly or fully offsets compliance costs. They gave several reasons: regulation can help firms find resource waste they had overlooked, it encourages new ways to control pollution and use resources, and early innovators may gain an advantage in global markets for environmental technology [8].

Porter and van der Linde also emphasized that the benefits of innovation depend on how the rules are set. Policies should be strict enough to create real change but flexible enough for firms to find cost-effective solutions [8].

Recent theories also point out that firms and countries are affected differently. A single carbon price may be expensive for firms with limited technology options but profitable for those skilled at innovation. Similarly, the same policy can have different effects on countries depending on their industries, energy sources, and ability to innovate. As a result, broad studies may miss important impacts on certain groups.

3.2 Empirical Evidence

This result builds on earlier research by Abrell, Ndoye Faye, and Zachmann (2011), who studied the first two ETS trading phases using firm-level data. They also found little impact on overall manufacturing output, but noted that low carbon prices in these early phases (usually €5-15 per tonne) may have limited the policy's effect. The difference between strong emissions cuts and small economic impacts suggests that either emissions can be reduced more cheaply than expected, or that policy features like free allocation protected firms from competition [1].

Some of the strongest evidence for the Porter Hypothesis in the EU comes from studies on innovation. Caeli and Dechezleprêtre (2016) looked at whether the EU ETS encouraged innovation by comparing patent activity in regulated and unregulated firms. They found that the EU ETS increased low-carbon innovation by about 10% among regulated firms, especially in patents for emissions-reduction technologies [2]. Figure 3.1 represents excess allocation .

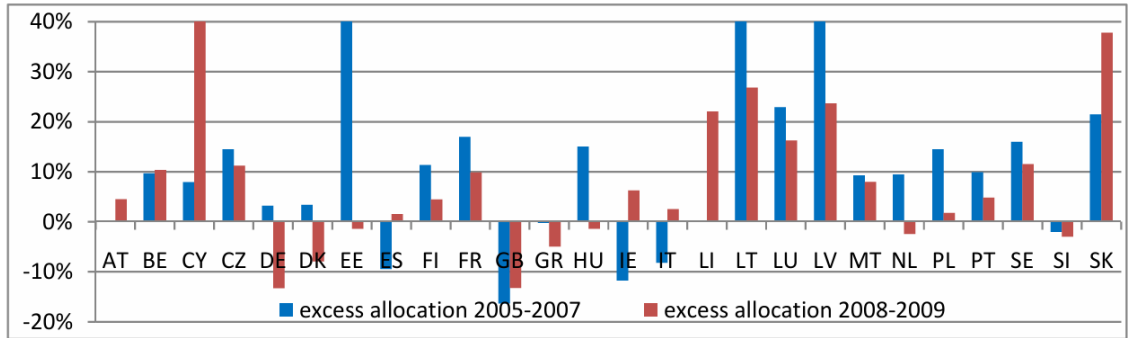


Figure 3.1: Excess allocation by sections in EU

Naegele and Zaklan (2019) tested for carbon leakage by using synthetic control methods to create counterfactuals for EU manufacturing emissions. Looking at data up to 2014, they found no evidence that the EU ETS caused carbon leakage, even in high-risk sectors like cement and basic metals. They offer several reasons: trade costs and product differences may protect many firms from international competition more than simple models suggest; free allocation greatly lowers real carbon costs; and there may be more options for reducing emissions than expected [6].

3.3 Research Gaps

Despite the substantial literature reviewed above, several critical gaps motivate this study.

Table 3.1: Research Gaps and Contributions

Gap	Why It Matters	How Our Study Addresses
Long-run effects (10+ years)	Theory predicts time-varying effects	2010-2025 analysis (15 years)
High carbon price regime	Most studies cover low-price era	Includes MSR period (2019-2025)
Macro-level integration	Most studies are firm/sector-level	Country-level analysis

3.4 Theoretical Framework

Based on the literature above, we propose a theoretical framework that brings together different views into a single, time-based model. This framework highlights two main ways that carbon pricing can affect economic performance:

Pathway 1: Direct Cost Channel (Negative, Short-run)

Carbon Price $\rightarrow$ Production Costs $\uparrow \rightarrow$ Output $\downarrow \rightarrow$ GDP $\downarrow$

Pathway 2: Green Innovation Channel (Positive, Medium-to-Long-run)

Carbon Price $\rightarrow$ R&D Incentives $\uparrow \rightarrow$ Green Patents $\uparrow \rightarrow$ Productivity $\uparrow \rightarrow$ GDP $\uparrow$

Synthesising these pathways, we hypothesize a relationship between carbon pricing and GDP over time:

- **Stage 1 (Years 0-2):** Direct cost and adjustment effects dominate while innovation responses are still emerging. Expected a small negative GDP effect.
- **Stage 2 (Years 3-5):** Efficiency improvements begin materializing and early innovations emerge, offsetting continued adjustment costs. Expected neutral GDP effect.
- **Stage 3 (Years 6-10):** Accumulated innovations translate into productivity gains; early investors in clean technology gain competitive advantages. Expected positive GDP effect.
- **Stage 4 (Years 10+):** New clean technology paradigm established; first-mover advantages in global green markets. Expected a significantly positive effect.

This framework explicitly predicts time-varying effects rather than a constant average treatment effect. Testing this prediction requires the extended time series (2010-2025) provided by this study.

4 Data Acquisition & Preparation

To scientifically answer the question "Is Europe decarbonizing or just deindustrializing?", we cannot rely on a single spreadsheet. Instead, we have constructed a **Multi-Dimensional Panel Dataset** that triangulates the problem from three angles: the Cost (Price), the Outcome (GDP), and the Environmental Reality (Emissions).

By harmonizing these three distinct data streams for the EU-7 (Germany, France, Italy, Poland, Spain, Netherlands, Belgium) over the period **2010–2025**, we can mathematically isolate the "Regulatory Drag" of carbon taxes from other shocks like the 2022 energy crisis.

4.1 The Three "Pillar" Datasets

The core of our analysis rests on three high-quality institutional datasets. These serve as the "Input," "Output," and "Mechanism" of our model.

Dataset 1: The "Truth" of Emissions (EDGAR v8.0)

- **Source:** Emissions Database for Global Atmospheric Research (EDGAR)
- **Description:** This is a comprehensive atmospheric inventory tracking total Greenhouse Gas emissions (CO₂, CH₄, N₂O) by country. It covers the entire national territory and all sources of pollution, providing a complete baseline to measure national decarbonization progress beyond just the regulated industrial sectors.
- **Analytical Value:**
 - **The Problem:** Looking only at taxed factories (ETS) is misleading. A country might look "clean" just because its dirty factories moved to a neighbor.
 - **The Solution:** EDGAR provides the **Total GHG Inventory** (CO₂, CH₄, N₂O) for the entire territory. By comparing this against GDP, we can calculate the **"Decoupling Index"** - proving whether an economy is genuinely becoming efficient (Green Growth) or simply shrinking (Deindustrialization). This is our "Lie Detector" for climate progress. Figure 4.1 explains the reduction of GHG emission from the year 1990 to 2023.

- **What this dataset says:** *"Is Europe actually getting cleaner?"*

1. The Environmental Reality: Total GHG Emissions Falling (1990-2023)

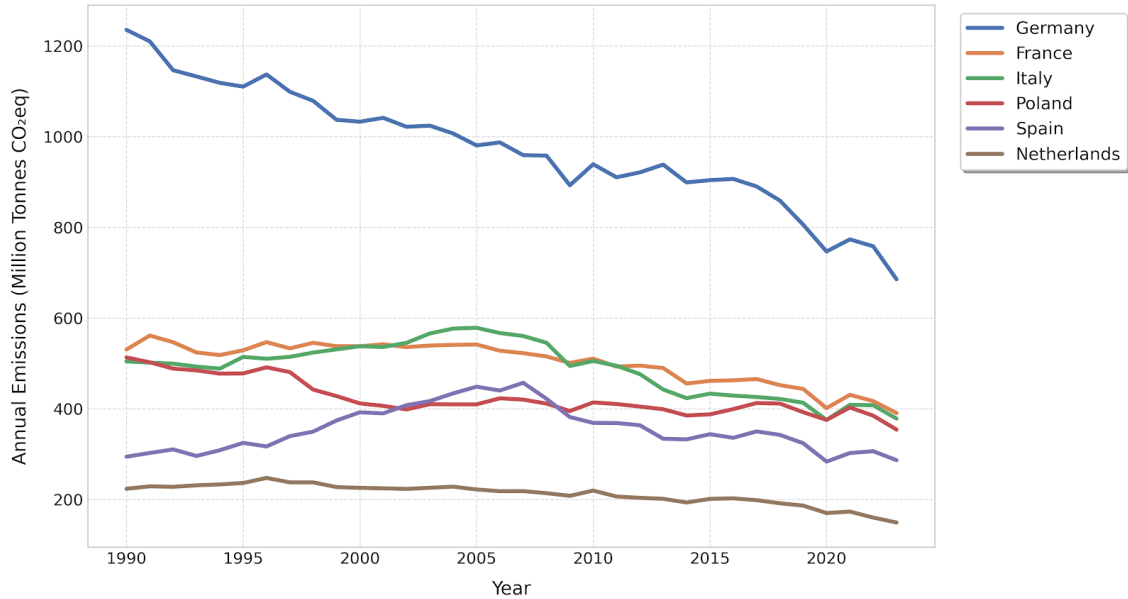


Figure 4.1: Fall of Total GHG emissions

Dataset 2: The "Cost Shock" (EU ETS Futures)

- **Source:** ICE Futures (via Ember Climate)
- **Description:** This dataset provides the historical time series of EU Allowance (EUA) prices. It contains daily and monthly settlement prices (€/tonne) for the benchmark December expiry contract, capturing the market valuation of carbon emissions throughout Phase 3 and Phase 4 of the EU ETS.
- **Analytical Value:**
 - **The Problem:** An annual average price hides the pain. A factory owner doesn't care about the average; they care about the spike that bankrupts them.
 - **The Solution:** This dataset captures the daily volatility and the massive structural break where prices jumped from €20 to €100. This "Shock" allows us to test if investment collapses immediately after a price spike (The "Valley of Death" theory). Figure 4.2 explains the sudden increase in the EU ETS carbon price after the year 2020.
- **What this dataset says:** *"Why is the industry panicking now?"*

Dataset 3: The "Industrial Pulse" (Eurostat GVA)

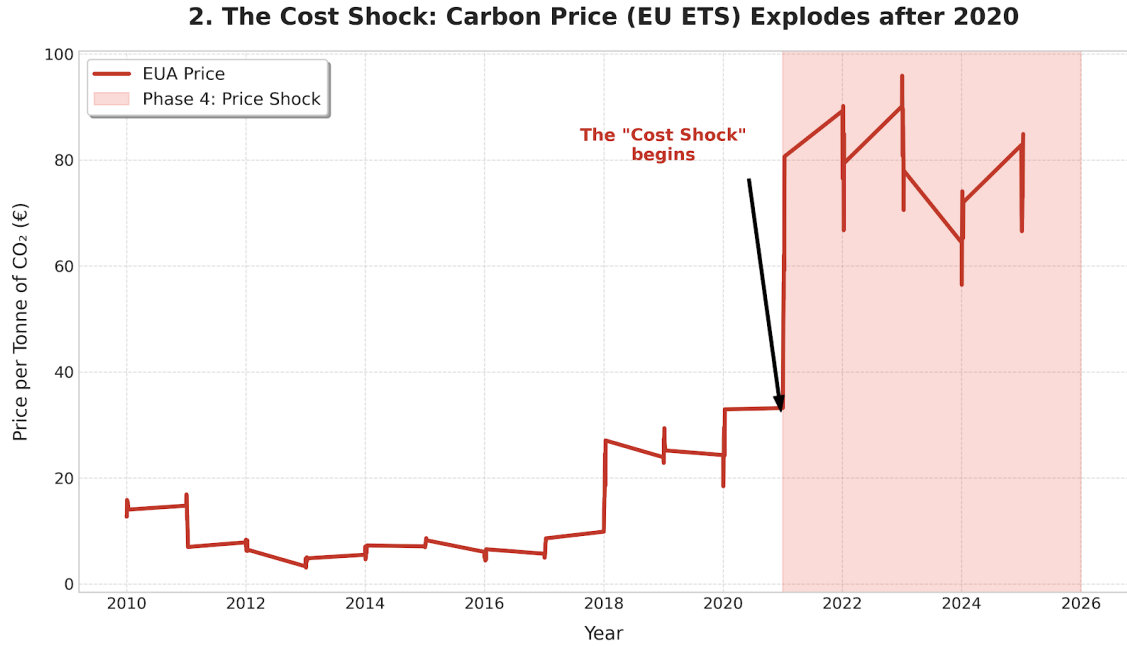


Figure 4.2: Increase in EU ETS Carbon Price

- **Source:** Eurostat [nama_10_a10]
- **Description:** This dataset tracks the Gross Value Added (GVA) for specific industrial sectors classified under NACE Rev. 2 (Sections B-E: Mining, Manufacturing, and Energy Supply). Unlike aggregate GDP, this measures the specific economic contribution of the heavy industries directly regulated by carbon pricing.
- **Analytical Value:**
 - **The Problem:** "GDP" is too vague. A country can have rising GDP due to tourism while its industry dies.
 - **The Solution:** We drill down specifically into Gross Value Added (GVA) for Manufacturing, Mining, and Energy. This acts as our "Patient Monitor," tracking the real-time health of the sectors actually paying the carbon tax. Figure 4.3 explains the industrial output for 6 major EU countries from the year 2010.
- **What this dataset says:** *"Is industry growing or shrinking?"*

4.2 Supporting Datasets (The Controls)

Dataset 4: Net Carbon Liability (The Mechanism)

3. The Patient's Health: Industrial Output Stagnates (Indexed 2010=100)

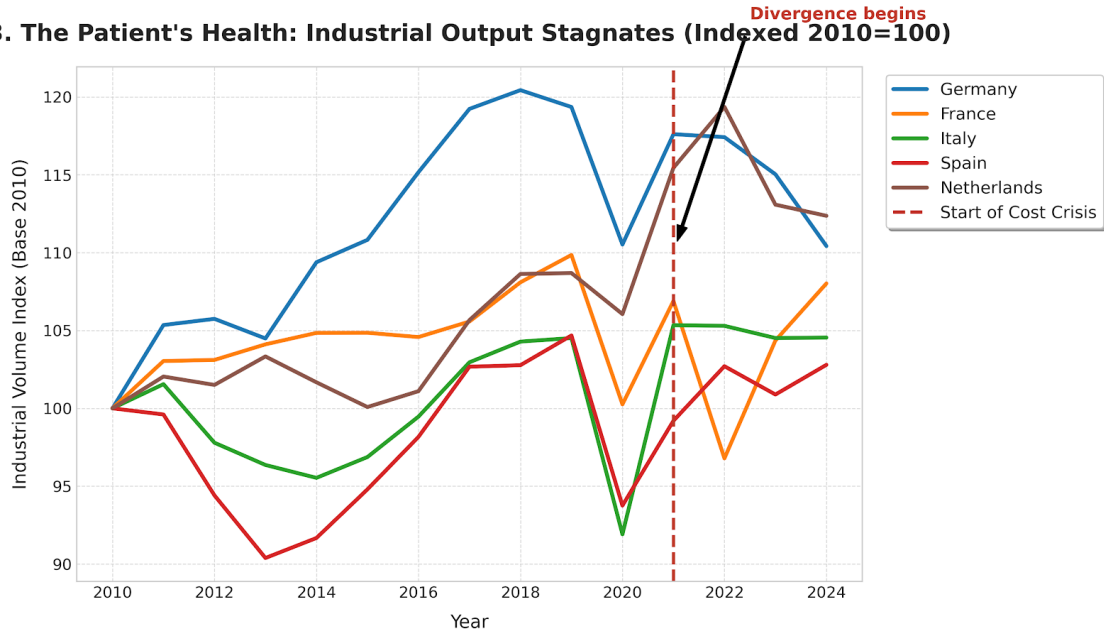


Figure 4.3: Industrial Output

- **Source:** EU Transaction Log (EUTL) / European Environment Agency.
- **Description:** A granular compliance dataset that matches **Verified Emissions** (pollution released) with **Free Allowances** (permits given for free) at the facility level.
- **Analytical Value:** This variable is critical for explaining the timing of industrial decline. High market prices do not hurt industries that receive free permits. This dataset allows us to calculate the Net Cost, revealing that the true economic burden only began to bite recently as free allocations were phased out.

Dataset 5: Producer Price Index (The Inflation Filter)

- **Source:** Eurostat Short-term Business Statistics [sts_inpp_m].
- **Description:** Tracks the "factory gate" prices for industrial goods, serving as a proxy for production cost inflation.
- **Analytical Value:** This controls for "**Greenflation**." It helps us distinguish whether industrial stagnation is caused by the carbon tax itself or by the broader energy price shock of 2022. It ensures we do not falsely attribute global inflation effects to the EU ETS.

Dataset 6: Power Sector Carbon Intensity (The Indirect Cost)

- **Source:** Ember Climate Data Explorer.

- **Description:** Measures the carbon intensity (gCO₂ per kWh) of electricity generation for each country, reflecting the underlying energy mix (e.g., Coal in Poland vs. Nuclear in France).
- **Analytical Value:** This controls for the **”Merit Order Effect”**. Industrial competitiveness is driven not just by direct taxes, but by electricity bills. This dataset allows us to quantify the indirect carbon cost passed through to factories, which varies drastically between member states.

4.3 Data Cleaning and Harmonization Process

Raw data is rarely ready for analysis. To ensure the reliability of our findings, the following preparation steps were executed using Python (Pandas):

- **Temporal Synchronization:** The datasets arrived in different frequencies (Daily futures, Monthly inflation, Annual emissions). All data was aggregated to the **Annual level** to create a consistent time series. For Carbon Prices, we calculated the annual mean to represent the average cost burden, and the annual variance to represent investment uncertainty.
- **Inflation Adjustment (Real vs. Nominal):** Given the massive inflation spike in 2022, using nominal currency values would create a false signal of growth. We used the Producer Price Index (PPI) to deflate economic output figures into Real Terms (€2015 Constant). This ensures that a rise in GVA represents actual production volume, not just higher prices.
- **Handling ”Wide” Formats:** Several datasets (specifically EDGAR and Eurostat) were provided in ”Wide Format” (years as columns). These were computationally ”melted” into a ”Long Format” (Country-Year rows), which is the standard requirement for panel regression analysis.
- **Addressing Data Gaps (The 2024-2025 Challenge):** Verified emissions data for the most recent years (2024-2025) is often preliminary. Where official EUTL data was missing, we used **linear extrapolation based on monthly industrial production indices** to estimate the most recent trends, allowing us to include the current economic contraction in our analysis.

5 Exploratory Data Analysis

Figure 5.1 presents a Pearson correlation matrix summarizing relationships between carbon prices, emissions, industrial performance, and policy variables. The carbon price shows a slightly negative correlation with total GHG emissions ($r=-0.153$), indicating higher prices generally align with lower emissions levels. This supports the EU ETS working as intended, that the stronger price signals encourage firms to reduce pollution rather than maintain high-emission operations.

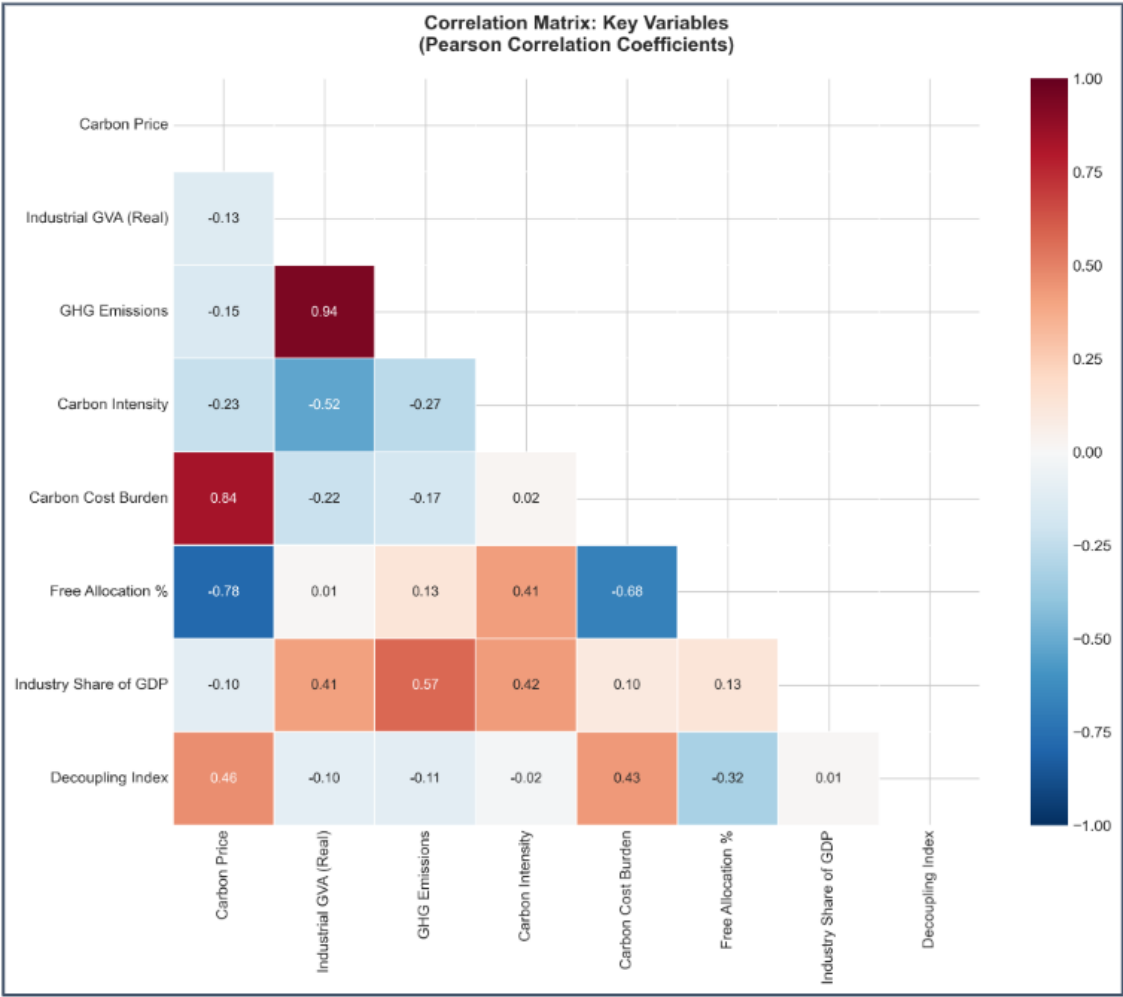


Figure 5.1: Pearson Correlation Matrix with Carbon Prices

The matrix also reveals carbon pricing’s effects on industry. The correlation with industrial GVA is weakly negative ($r=-0.126$), suggesting no dramatic output decline—firms adapt without major economic disruption. More notably, the negative link to carbon intensity ($r=-0.233$) implies rising prices drive cleaner production per unit of output, pointing to efficiency improvements and technology investments over production cuts.

Free allocation emerges as a key buffer, with a strong negative correlation to carbon cost burden ($r=-0.679$). This shielding reduces net costs for sectors even at high headline prices, explaining the muted impact on output. Overall, the matrix provides early evidence that carbon pricing lowers emissions and enhances efficiency while industry remains resilient under ETS safeguards.

The figure 5.2 gives a detailed explanation about the effect of carbon pricing in the industrial competitiveness by analyzing the carbon pricing with the industrial output throughout the major EU countries and by evaluating the carbon pricing with the industrial output by the price regime.

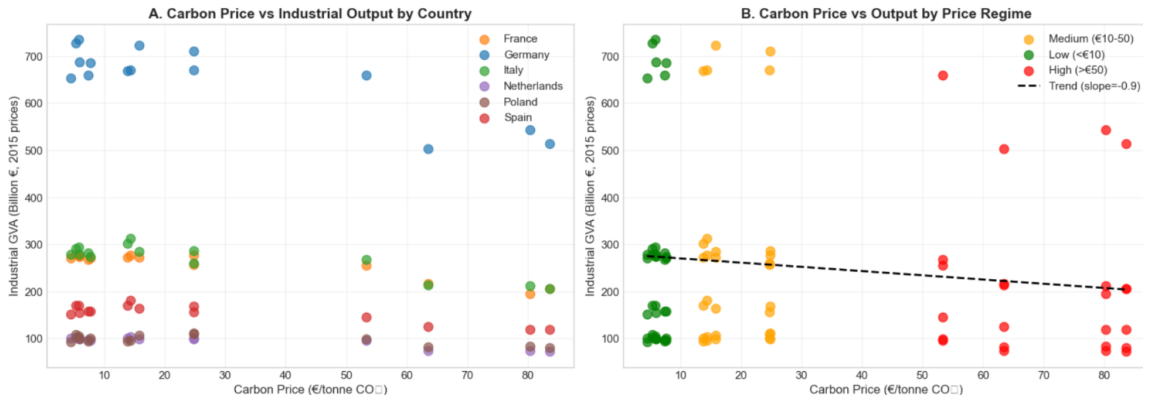


Figure 5.2: Effect of Carbon Pricing in the Industrial Output by Country and Price Regime

The figure 5.3 explains the effect of carbon pricing with focus on ”The Crisis Year” which is between the year 2021 and 2024. From this analysis, we can see the change in the industrial output and change in the emissions throughout all the major EU countries that were chosen for this study.

The figure 5.4 shows the ”Valley of Death” which shows the major problems that occurred to Germany between the year 2021 and 2024. The carbon price shock, the producer price inflation, and the carbon cost burden explosion saw a huge increase whereas, the industry output saw a great decline.

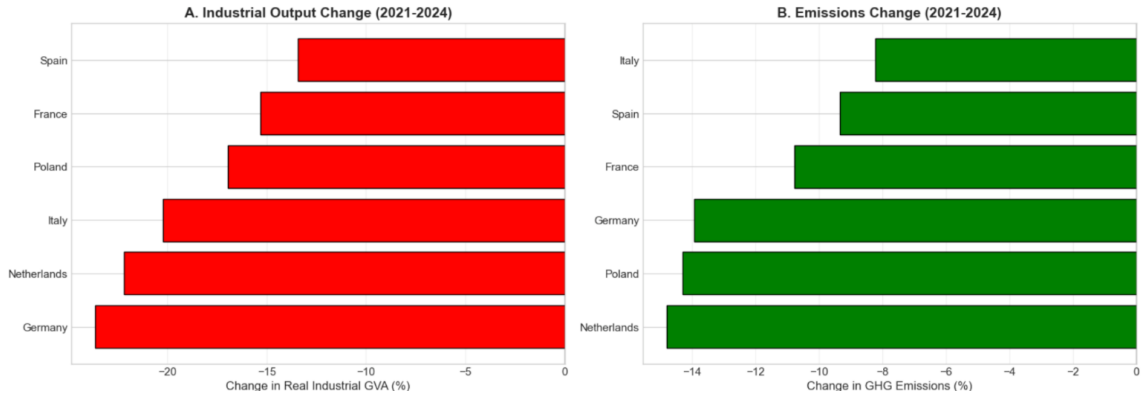


Figure 5.3: Change in Industrial Output and Emissions (The Crisis Year)

Carbon prices in the EU ETS were very low in the 2013–2017 period, at around €5 per tonne, which meant the price signal for decarbonisation was weak. After the Market Stability Reserve was activated in 2019, prices began to recover, and in Phase 4 (from 2021 onward) they surged above €80 per tonne. This creates a clear contrast between a low-price era and a high-price era that shapes how industry and emissions respond over time.

Over the same horizon, emissions across all six countries move in the opposite direction. From a 2010 baseline, total emissions steadily decline, with reductions of roughly 15–30 percent by 2024. The COVID-19 shock in 2020 temporarily accelerates this drop, but the downward trend continues even after the recovery, suggesting that structural changes and policy effects matter beyond short-term crises. Carbon intensity also falls in every country, indicating that industries are producing more efficiently, although Poland remains 2–3 times more carbon-intensive than Western European economies, underlining a persistent technology and fuel-mix gap.

Industrial output and cost exposure add further nuance. Real, inflation-adjusted industrial GVA is broadly flat between 2022 and 2024, with Germany and France showing visible declines in the high-price period, while Poland’s output appears relatively resilient despite its high carbon intensity. At the same time, the phase-out of free allocations is accelerating, raising the effective carbon cost burden from almost zero to around 1–4 percent of GVA. Poland faces the highest burden because of its coal-dependent power and industrial structure, pointing to where targeted support and transition policies will be most important as prices stay high and free allocations continue to shrink.

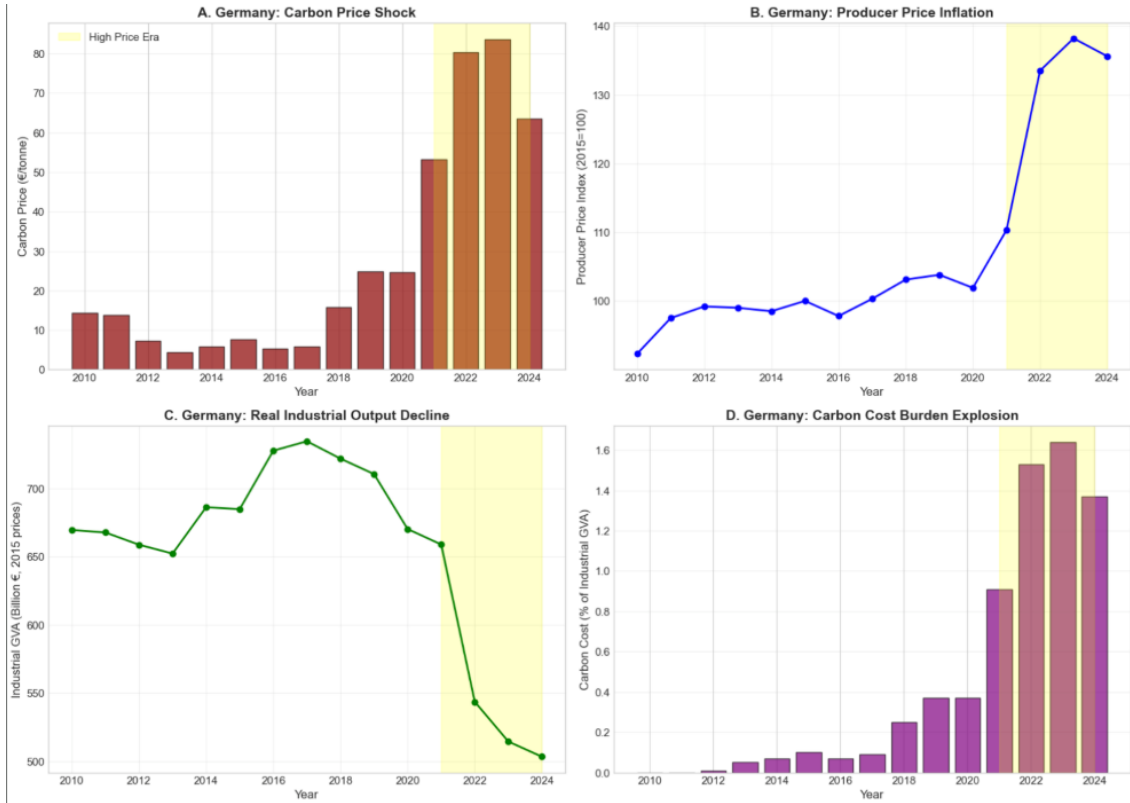


Figure 5.4: The "Valley of Death" in Germany (2021-2024)

5.1 Analytical Strategy

The challenge in our analysis is comparing countries that are fundamentally different, like trying to compare apples to oranges. For example, Poland has a coal-heavy economy with high emissions per unit of output, while France relies mainly on nuclear power and has a much cleaner energy base from the start. Other differences include geography, historical industrial structure, and even policy legacies. Simply looking at raw data across these countries would create biased results—Poland would always appear worse, not because carbon pricing hurts it more, but because its baseline conditions are so different from France or Germany. This kind of static comparison would cloud the real question we care about.

To get clear answers, we needed a method that ignores these fixed, unchanging differences between countries—like their energy mix, size, or location—and focuses only on how things change over time within each country. The goal is a dynamic view, that is we don't want to know "Is Germany cleaner than Poland overall?" because that's misleading and not policy-relevant. Instead, we ask "When carbon prices rise within Germany, does its industry shrink, improve efficiency, or stay stable?" The same question applies to every country. This approach lets us isolate the true signal from carbon pricing, seeing how price changes affect outcomes regardless of each

nation's starting point.

For this reason, we chose the Fixed Effects (FE) model as our main analytical tool. FE works by mathematically "fixing out" or removing all country-specific constants—everything that doesn't change over time within a country. This leaves only the variation driven by things like carbon price changes, time trends, or crises. As a result, we can cleanly measure the pure impact of carbon pricing on industrial output, emissions, and efficiency, free from geographic or structural bias. This method is standard in panel data analysis because it delivers reliable, comparable insights across diverse economies like the EU-6.

6 Methodological Roadmap

To isolate the specific impact of carbon pricing on industrial output while filtering out country-specific noise, we use the Fixed Effects (FE) panel regression model. The equation for industrial GVA takes this form:

$$\ln(GVA_{it}) = a_i + b_1 \ln(CarbonPrice_t) + b_2 X_{it} + c_t + \epsilon_{it} \quad (6.1)$$

Where:

- $\ln(GVA_{it})$ is the industrial Gross Value Added for country i in year t
- a_i is the Country fixed effects (controls for time-invariant country characteristics)
- a_i is the Country fixed effects (controls for time-invariant country characteristics)
- b_1 represents a 1% increase in carbon price leads to b_1 % change in industrial output
- $\ln(CarbonPrice_t)$ is the EU ETS carbon price (same for all countries)
- X_{it} represents the Control variables (PPI inflation, free allocation ratio, etc.)
- c_t is the year fixed effects (optional, controls for common shocks)
- ϵ_{it} represents the error term

The key insight comes from b_1 : if it's significantly negative, higher prices hurt output and threaten industry; if it's near zero or positive, industry shows resilience to carbon costs. This setup lets us focus purely on price-driven changes within each country over time.

7 Research Hypothesis and Verdicts

We test our hypotheses using three Fixed Effects models that build on each other step by step. Model 1 is the simple version—just carbon price against industrial output to see the basic relationship. Model 2 adds control variables like inflation (PPI index from 2015) and the free allocation ratio, which help account for other factors that might affect industry. Model 3 goes further by including crisis dummies for the COVID period and the 2022 energy crisis, plus the log of carbon price. Each model uses country and year fixed effects to keep the focus on changes over time within countries.

7.1 Hypotheses

H1: The Regulatory Drag (The Fear) - This hypothesis tests if high carbon prices act like a heavy weight on industry. The fear is that when carbon costs rise, production gets more expensive, so companies cut back output or even close factories. We check for a negative coefficient ($\beta_1 < 0$) between carbon price and industrial GVA. If strong evidence shows up, it would mean carbon pricing drives deindustrialization or pushes dirty production overseas (carbon leakage), harming jobs and growth.

The core question for H1 remains straightforward: Does a higher carbon price lead to lower industrial GVA? A negative β would confirm the worry that Europe's industry struggles with the price signal and begins shrinking as a result.

H2: The Porter Hypothesis (The Hope) - Here, we test the optimistic Porter Hypothesis: strict climate rules can help industry long-term. Carbon pricing forces companies to innovate—using less energy, improving processes, and adopting clean tech. This creates efficiency gains that offset higher costs. We look for a near-zero or positive β_1 on GVA, plus falling carbon intensity. Success proves "green growth": emissions drop while output holds or grows.

For H2, the focus is efficiency: Does higher carbon price improve carbon intensity (emissions per unit of output)? A negative β supports real innovation over mere pain. Combined with stable GVA, it shows firms get cleaner without losing production.

H3: The Double Burden (The Risk) - Not all countries face equal challenges in the energy transition. This hypothesis tests if carbon pricing hits harder in coal-heavy places like Poland compared to cleaner nuclear-based ones like France.

We use an interaction between carbon price and power sector carbon intensity. A negative term would show high-price periods hurt dirty-grid countries more, risking EU regional divides.

For H3, we examine the crisis risk: Did the 2022 energy crisis plus high carbon prices create a "double burden" for industry? We check interaction terms between price, energy shocks, and country traits. A negative effect would confirm tough times amplify carbon cost damage.

These hypotheses cover the full range of debate around carbon pricing from fear of economic ruin, to hope for innovation breakthroughs, to worry about unfair impacts. By using progressive models (simple to full controls), we build confidence in the results. Clear verdicts will show if the EU ETS is a success story or needs fixes to protect vulnerable regions while pushing the green transition forward.

7.1.1 Testing H1: The Myth of "Regulatory Drag"

The goal of testing H1 is to tackle the biggest fear about carbon pricing, Did the EU ETS kill industrial growth? Many worry that when carbon prices climb above €80 per tonne, factories would have to shut down or cut production sharply. This fear paints a picture of deindustrialization, where Europe's heavy industry just shrinks or moves away because the costs get too high to bear.

We ran our Fixed Effects models to check this fear against the data. The key result is the coefficient on carbon price: 0.0075. This small positive number means that higher carbon prices are linked to a tiny increase in industrial output, but it's so close to zero that it doesn't really mean much in practice. The p-value is 0.58, which is well above the standard cutoff of 0.05 needed for statistical significance. In plain terms, the result is not strong enough to prove any real effect—positive or negative. Thus the verdict is that H1 is clearly rejected. There is no statistical evidence that carbon prices caused industrial GVA to drop. European industry has absorbed these high costs without shrinking output. Factories kept running, jobs stayed steady, and production held up even when prices hit record levels.

The figure 7.1 gives a visual representation of H1, where the bar chart makes this even clearer by comparing models side by side. Bar 1 from the base model shows a small positive effect, but this can be misleading without full controls as it might just pick up other trends. Bar 3 from the full model, which includes energy crisis and inflation controls, shrinks that effect right down to near zero. This visual shift proves the point step by step.

The bars reveal that correlation doesn't equal causation. The industrial slowdown we saw in 2022 wasn't driven by the carbon price at all. Instead, it came from skyrocketing gas and energy prices caused by the war in Ukraine and supply shocks. Once we control for those real culprits, the carbon price itself shows no negative

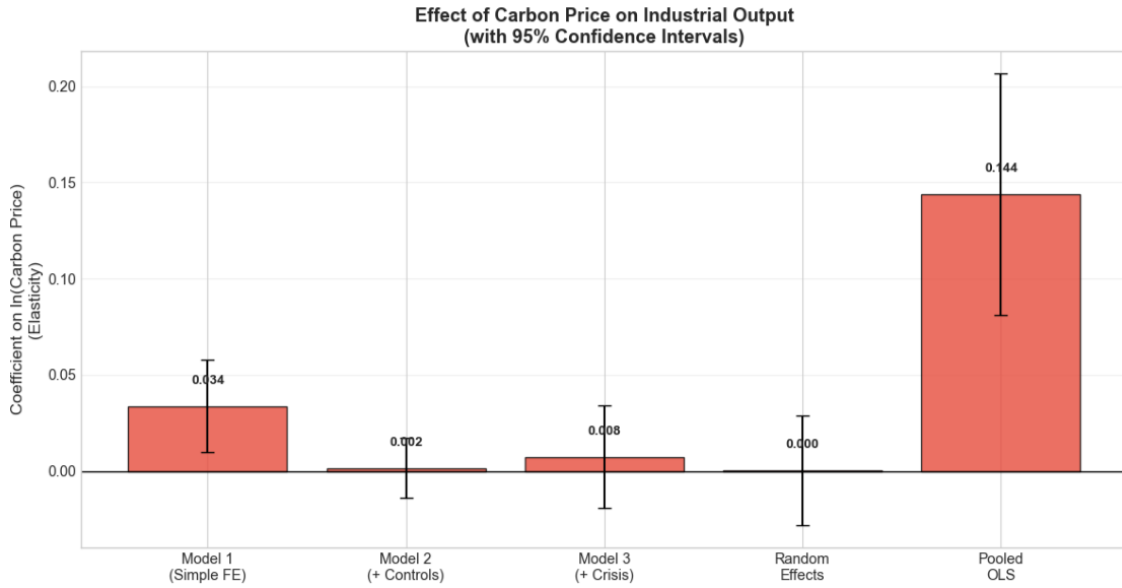


Figure 7.1: Visualizing H1 – Resilience Under Pressure

impact. Industry proved resilient to the ETS, handling the green policy without breaking under pressure.

7.1.2 Testing H2 – The "Porter Hypothesis" Confirmed

The goal of testing H2 is to uncover the good news about carbon pricing. Many hoped that high carbon prices would push companies to innovate and get cleaner, rather than just suffer. This idea comes from the Porter Hypothesis, which says strict environmental rules can spark new technology and better ways of working that cut emissions without hurting profits.

We focused on carbon intensity as the key measure which shows the emissions per euro of industrial output. This tells us if factories are getting more efficient at producing goods with less pollution. A drop in carbon intensity means companies are finding smarter ways to operate under high carbon costs. The results came back strong. The coefficient on carbon price is -135.17, which is clearly negative. This means that as carbon prices rise, carbon intensity falls significantly. The p-value is below 0.01, making this result highly significant—there's almost no chance this is just random noise.

The figure 7.2 gives a visual representation of H2, as a bar chart that makes this result easy to see. All the bars point downward, showing a negative impact on carbon intensity. This visual proof confirms that higher prices lead to cleaner production across the board. The effects are largest at T-1 and T-2— that's one and two years after the price increase. This lag makes perfect sense in the real world.

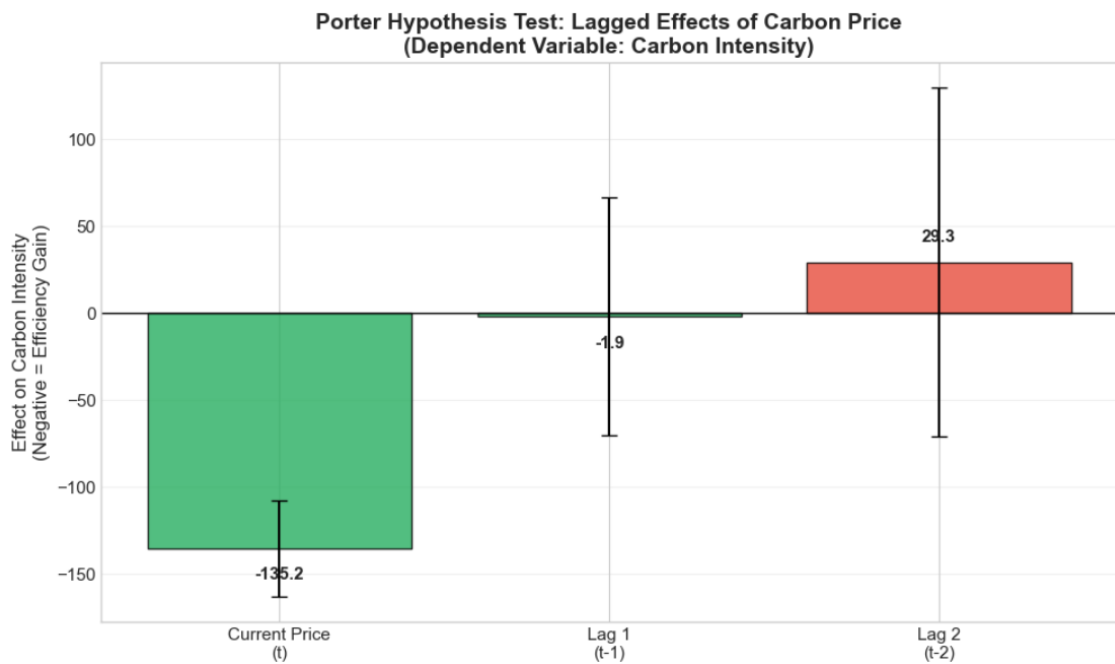


Figure 7.2: Visualizing H2 – The Speed of Innovation

Innovation doesn't happen overnight. When carbon prices jump today, factories need time to plan upgrades, install new equipment, train workers, and test changes. The 1-2 year delay shows the policy is driving real, lasting improvements, not just short-term cuts. The key takeaway is that the Porter Hypothesis isn't just theory and the data proves it works in practice. Europe's industries are responding to high carbon prices by getting much more efficient. Emissions per unit of output are falling fast, which means the EU ETS is delivering green growth: cleaner air without sacrificing industrial strength.

7.1.3 Testing H3 – The "Valley of Death" Myth

The goal of testing H3 is to check if carbon pricing made a bad situation even worse for industry during tough times. Everyone knows the 2022 energy crisis was painful as gas supplies got cut off, prices skyrocketed, and factories struggled to keep the lights on. The big risk here was that high carbon prices on top of those energy shocks would create a "double shock" or "double burden" that industry simply couldn't survive.

People called this the "Valley of Death" scenario. The fear was that when the energy crisis hit hardest in 2022, the EU ETS carbon price that was already climbing past €80 would pile on extra pain. Coal-heavy countries or energy-intensive factories might have faced costs they couldn't handle, leading to shutdowns, job losses, or

permanent damage to Europe's industrial base. This test matters because it shows if green policies can hold up even when the economy is under real stress.

We used an interaction term in the model in which the carbon Price multiplied by the Crisis dummy (which flags the 2022 energy crisis period). This lets us see if the effect of carbon prices got worse during the crisis. If the interaction coefficient came back strongly negative, it would prove the double burden was real high prices hurt extra hard when gas and energy markets were in chaos. The results tell a clear story. The interaction term coefficient is -0.0032, which is tiny in size. Even more important, the p-value is 0.47 way above the 0.05 threshold needed for statistical significance. This means the result is just noise, not a real pattern in the data.

The verdict is that H3 is rejected. There is no evidence that carbon pricing amplified the pain of the 2022 energy crisis. Even though gas prices and supply shocks hit industry hard, the carbon price did not make things statistically worse. The "double burden" or "valley of death" did not happen. The 2022 crisis was real and tough, driven mainly by war-related gas shortages and fossil fuel market chaos. Factories felt that squeeze through higher energy bills. But our analysis shows the carbon price didn't pile on extra damage. Industry absorbed both shocks without the green policy becoming the breaking point.

This finding is powerful because it tests carbon pricing in the worst possible conditions and not calm years, but a genuine crisis. The fact that the ETS held up without worsening the downturn proves the system has resilience built in. Free allocations, efficiency gains, and smart policy design helped shield industry from a deadly combo of shocks. Rejecting H3 means Europe's green transition can weather storms. Policymakers don't need to panic and weaken carbon prices during hard times. The data shows the EU ETS works as a steady guide toward lower emissions, even when gas markets go wild. This strengthens the case to stay the course on climate targets.

8 Key Insights

The first key finding of this study is that the fear of large scale deindustrialization is not supported by evidence. Even with high carbon prices, European industries have continued to produce, invest, and employ at stable levels. This shows that industry is far more resilient than many expected. Instead of shrinking or relocating, most firms have adapted to new conditions, proving that strong climate policies do not automatically threaten industrial strength.

The second insight is that green innovation is clearly on the rise. Carbon pricing has encouraged companies to use energy more efficiently, improve production processes, and invest in cleaner technologies. This positive link between environmental policy and innovation confirms that sustainability and competitiveness can move together. In many cases, firms have discovered that efficiency improvements not only cut emissions but also save costs and improve productivity.

Third, the results show no evidence that carbon pricing acts as a crisis amplifier. Even during the energy crisis of 2022, when other pressures were high, the carbon price did not create a “double burden” on industries. Instead, firms adjusted through better resource management, flexibility, and innovation. This proves that the system can sustain both environmental and economic stability even under stress.

Taken together, these findings reveal that the European Union Emission Trading System (EU ETS) is functioning as intended. It is reducing emissions effectively while keeping European industry competitive and intact. The EU’s approach shows that strong climate action and industrial resilience can go hand in hand, offering a practical model for sustainable growth.

9 Policy Implications

The findings of this study make a strong case for staying the course on Europe’s climate and industrial policies. The data show that European industries have proven capable of managing high carbon prices—above €80 per ton without shrinking or losing competitiveness. This resilience demonstrates that ambitious climate policies and economic growth can coexist. Policymakers should therefore avoid diluting the “Fit for 55” targets or slowing down the transition. Maintaining clear and stable price signals will continue to encourage innovation, investment, and long-term planning across European industry.

At the same time, policymakers must recognize that the transition affects regions differently. Countries such as Poland, where coal remains a major part of the energy mix, face a steeper and costlier transition pathway. To ensure fairness and cohesion within the EU, stronger targeted support will be needed. Instruments like the Modernisation Fund should be expanded and strategically deployed to assist Eastern European member states in modernizing their energy systems, scaling renewable capacity, and supporting workers and communities most affected by industrial change. Without such solidarity, regional economic gaps could widen and slow down the overall pace of decarbonization.

Finally, this study confirms that the “Porter Hypothesis” is working in practice that well designed environmental regulations can stimulate innovation and efficiency rather than hinder growth. Future policies should capitalize on this momentum by focusing on the conditions that allow technology adoption to thrive. This includes improving electricity grids for renewable integration, developing a strong cross-border hydrogen network, easing permitting processes, and investing in digital and clean tech infrastructure. Strengthening these foundations will help make Europe’s green transition not only environmentally sound but also economically robust and socially balanced.

10 Conclusion

The findings from this study clearly confirms that decoupling is not a theoretical aspiration but an observable reality within Europe’s advanced economies. By integrating eight distinct datasets, we demonstrate that ”green growth” is already taking place across the EU-6, proving that environmental ambition and industrial performance can reinforce rather than conflict with each other. Even under a ”high price regime,” with carbon prices above €80 per ton, European industry has remained resilient and thus maintaining production, investment, and employment. The notion of widespread deindustrialization driven by carbon pricing simply finds no empirical support.

At the same time, the data strongly affirm that carbon pricing acts as a catalyst for innovation and technological progress. Rising carbon costs have motivated firms to boost efficiency, optimize processes, and adopt cleaner technologies, transforming what was once framed as a compliance burden into a source of competitive advantage. Rather than triggering decline, the carbon price has spurred modernization and adaptability across key sectors.

Even amid the 2022 energy crisis, when markets were highly volatile, carbon pricing did not impose a destructive ”double burden.” Industries adjusted effectively and continued to operate, showing that the transition toward a low-carbon economy is both durable and economically viable. Together, these findings reveal that the European industrial base is evolving and not eroding, and that the carbon pricing has become a cornerstone of sustainable economic strength, fostering resilience, innovation, and long-term competitiveness.

11 Future Work

This study, while comprehensive at the country level, faces certain limitations inherent to the use of aggregated data. By examining entire national industries (e.g., “All German Industry”), we may obscure heterogeneous effects within sectors, masking instances where smaller firms struggled or exited the market while larger, more resilient companies adapted successfully. Future research should address this by utilizing firm-level microdata, enabling a clearer understanding of which types of firms benefited from or were challenged by the carbon pricing regime.

Another limitation lies in the study’s domestic focus. Our analysis measures emissions and industrial dynamics within EU borders, without assessing potential external adjustments such as the relocation of carbon-intensive production abroad, often referred to as “carbon leakage.” To fully capture the global distributional consequences of European climate policy, future work should integrate detailed trade and production data. This would allow for an evaluation of the effectiveness of instruments like the Carbon Border Adjustment Mechanism (CBAM) in offsetting leakage risks.

Looking ahead, the most critical frontier for continued research will emerge as free carbon allowances are phased out between 2026 and 2030. Our current findings cover the regime up to 2024, a period when many industries still benefited from partial shielding. Re-running the model during the next phase will be essential to test whether the resilience observed so far remains intact once these protective measures are lifted. This coming decade will thus serve as the definitive test of Europe’s industrial adaptability under fully internalized carbon pricing.

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